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Magnetic Cilia and Hair-Follicle Architecture Enabled Dual-Mode Bionic E-Skin for High-Sensitivity and Wide-Linear-Range Tactile/Touchless Perception

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Pressure/magnetism dual-mode sensors for tactile/touchless perception are desired to adapt multifunctional applications. However, it remains a challenge to possess high sensitivity and a wide linear response range in both modes. Here, a bionic e-skin with magnetic cilia and hair-follicle architecture (MCHFA) is proposed to achieve high-sensitivity and wide-linear-range pressure/magnetism dual-mode perception (0–26 kPa: $S = 2.72$ kPa^{-1}, $R^2 = 0.99$; 26–400 kPa: $S = 0.60$ kPa^{-1}, $R^2 = 0.99$; 10–171 mT: $S = 2.8$ T^{-1}, $R^2 = 0.94$), where the high sensitivity is mainly dominated by the easy deformation of cilia and the wide linear range is attributed to the gradient compressibility of the synergistic architectures of the cilia and the hair-follicle. Additionally, high resolutions of pressure (0.20%) and magnetism (11.0%) are obtained due to the wide-linear-range high sensitivity in both modes. These excellent characteristics in dual modes endow the MCHFA-based sensor multifunctional applications in health monitoring and human–machine interfaces, such as a finger bending game for the rehabilitation training and the prevention of senile dementia, and a noncontact coded lock taking advantage of less risks in cross infection and personal information leakage.

1. Introduction

With the growth of the Internet of Things and artificial intelligence, sensors that can convert external excitations into electrical signals are becoming more and more crucial in smart lives.[1–6] However, these devices usually work independently and can only perform a single type of information specified for monitoring. These sensors can be classified as contact or non-contact depending on how the signals are transmitted. The development of dual-mode sensors for both tactile perception (e.g., strain[7,8] and pressure[9,10]) and noncontact sensing (e.g., temperature,[11,12] humidity,[13,14] light[15,16] and magnetic field[17,18]) is crucial given the variety of human needs and the growing complexity of the workplace. When noncontact interaction is highly desired, the pressure/magnetism responsive sensor, a representative dual-mode sensor, can not only enable the real-time collection of physiological signals of interest for human–machine interfaces and health monitoring via direct mechanical stimuli perception, but it can also respond quickly and sense distance-dependently to external stimuli, preventing a variety of potentially dangerous situations, such as bacterial infection from mechanical touch.[13,19–23]

With continually improved performance toward prospective applications, the pressure/magnetism dual-mode sensor has been the subject of intensive research recently.[13,19,22–29] With a high signal-to-noise ratio, the high sensitivity makes it possible to precisely identify even tiny pressure differences. Liu et al. propose a self-powered liquid-solid interface ferrofluid-based triboelectric tactile sensor with an instantly formed spike-shaped microstructure.[25] This sensor's high spike microstructure, low ferrofluid Young's modulus, and effective solid-liquid interface contact-electrification allow it to achieve an ultrahigh sensitivity of 21.48 kPa^{-1}. However, the linearity range of this ultrahigh sensitivity is only within 2.5 kPa. The large linearity range can make data processing and conversion easier while maintaining high-pressure resolution over a wide pressure spectrum. To achieve gradient compressibility, Ji et al.[26] created a hybrid dielectric consisting of a micro-dome array, a high-permittivity rough surface, and a low-permittivity micro-cilia array. This allowed for an ultra-wide linearity range up to 1000 kPa with a sensitivity of 0.314 kPa^{-1}. Consequently, it is still very difficult for dual-mode sensors in pressure responsive mode to have both a wide linear response range and high sensitivity at the same time.

Actually, the dual-mode sensors in magnetism-responsive mode also face the same challenge.[13,19–22,30–33] A magnetic tilting micro-pillar array was created by Zhou et al.[19] as the dielectric layer for a capacitive pressure/magnetism dual mode sensor. It has a relatively low sensitivity of 0.69 T^{-1} but a broad magnetic field linear response range of 25–400 mT. On the other

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hand, a 3D structured multifunctional sensor that connected a magneto whisker with a super flexible patterned skin film was able to obtain a high sensitivity of 1.52 T^{-1} in a narrow magnetic field linear response range from 25.5 to 40.6 mT.[20] The intended pressure/magnetism dual-mode sensors should therefore have a high sensitivity and a broad linear response range in both contact and noncontact modes in order to accommodate multifunctional application scenarios. Although existing tactile/touchless dual-mode sensors based on various work mechanisms (e.g., capacitive, triboelectric effect, and light waveguide, etc.) face the challenge in balance of high sensitivity and wide linear range for each mode (Table S1, Supporting Information),[6,34–36] the existing dual-mode sensor researches still provided a clue to achieving wide-linear-range high sensitivity in both modes, i.e., designing suitable hierarchical structures containing the magnetic materials.

Here, a bionic e-skin with magnetic cilia and hair-follicle architecture (MCHFA), i.e., a MCHFA-based sensor, is proposed to achieve high-sensitivity and wide-linear-range pressure/magnetism dual-mode perception, where the high sensitivity is mainly dominated by the easy deformation of cilia and the wide linear range is attributed to the gradient compressibility of the synergistic architectures of the cilia and the hair-follicle. Additionally, high pressure/magnetism resolution is obtained due to the wide-linear-range high sensitivity in both modes. The MCHFA-based dual-mode sensor also exhibits fast response/recovery time and remarkable stability. These excellent characteristics in dual modes endow the MCHFA-based sensor with multi-functional applications in health monitoring and human-machine interfaces, such as a finger-bending training game and a noncontact coded lock.

2. Results and Discussion

2.1. Pressure Sensor Design and Sensing Mechanism

Tactile perception, a feedback to mechanical stimulation on the skin, plays an important role in physiological activities. In addition to the skin itself, the cilia on the skin can also effectively transfer mechanical stimulation to the sensory nerve fiber around the hair follicle. Here, we design a bionic e-skin with magnetic cilia and hair follicle architecture (MCHFA), as shown in **Figure 1a**. The cambered surface of the hair follicle endows the magnetic cilia at different heights, and thus a gradient compressibility of the synergistic architectures, as shown in Figure 1b. The detailed preparation process of the MCHFA-based bionic e-skin is exhibited in Figure S1 (Supporting Information). From Figures S2 and S3 (Supporting Information), we can see most of the magnetic cilia shows conical structures with different height. Figure S3a,b (Supporting Information) suggested that the magnetic cilia around the hair-follicle usually posseess a higher protrusion height above the hair-follicle than those in the certer area due to the cambered surface of the hair-follicle. In this study, it is noticeable that the MCHFA-based bionic e-skin is a kind of piezoresistive pressure sensor with contact and noncontact response modes to external stimulation (Figure 1b). Existing tactile/touchless dual-mode sensors are usually based on the work mechanisms such as capacitive, triboelectric effect, and light waveguide, etc. (Table S1, Supporting Information),[6,34–36] There are few reports on the tactile/touchless dual-mode sensors based on the piezoresistive effect. The gradient compressibility of the magnetic cilia with different height on the hair-follicle architecture leads to a continuous deformation ability over a broad pressure range. As a result, the contact area between the pressure sensing layer and the electrodes, i.e., the number of conductive paths, keeps increasing in a wide pressure range (Figure 1b), guaranteeing high sensitivity over a wide pressure range. In contact mode, the external stimulation is mechanical. The cilia structure shows a high sensitivity to mechanical stimulation. When the MCHFA-based sensor is exposed to low pressure, the magnetic cilia with higher protrusion height above the hair follicle can be bended easily followed by the bending of the magnetic cilia with lower protrusion height, and then the bent cilia rapidly increase their contact area with the copper foil electrode, leading a quick decrease of the resistance $R_{\text{MC-anode}}$ between the magnetic cilia and the anode. At this stage, the total resistance R_{total} can be described as

$$R_{\text{total}} = R_{\text{MC-anode}} + R_{\text{MC}} + R_{\text{HFA}} + R_{\text{HFA-cathode}} \tag{1}$$

where R_{MC}, R_{HFA}, and $R_{\text{HFA-cathode}}$ are the resistance of the magnetic cilia and hair-follicle architecture, and the contact resistance between the hair-follicle architecture and the cathode, respectively. With the increasing pressure, the hair-follicle architecture begins deformation to compensate for the gradually saturated contact between the cilia with the electrode, resulting in a wide pressure response range. The total resistance is changed into

$$R_{\text{total}} = \frac{R_{\text{MC-anode}} \times R_{\text{HFA-anode}}}{R_{\text{MC-anode}} + R_{\text{HFA-anode}}} + R_{\text{MC}} + R_{\text{HFA}} + R_{\text{HFA-cathode}} \tag{2}$$

due to the co-compression of hair-follicle architecture and magnetic cilia, where $R_{\text{HFA-anode}}$ is the contact resistance between the hair-follicle architecture and the anode. Finite element analysis (FEA) further confirmed the gradient compressibility of the synergistic architectures of the magnetic cilia with different height and the hair-follicle (Figure 1b). Although there are some similar biomimetic structures with the magnetic cilia,[19,21,27] such as microneedle array and tilted magnetic micropillars, they usually show a uniform cilia height, leading to a quickly saturated response to external pressure. Therefore, the sensors only based on the magnetic cilia or similar biomimetic structures usually show small linear response ranges with low sensitivities, as shown in Table S2 (Supporting Information). The synergistic effect of magnetic cilia with different height and hair-follicle architecture provides a novel approach to achieving high sensitivity and wide linear response range. In terms of the non-contact mode, carbonyl iron powder (CIP) in the cilia provides a non-contact responseability to external magnetic field instead of mechanical stimulation. The dual modes of the MCHFA-based sensor endow it multi-functional applications in daily life (Figure 1c,d). On one hand, a finger-bending training game was designed by attaching the sensors to the finger joints to monitor the finger-bending exercise, which can be used to the rehabilitation training for finger hurt or hemiplegia and the prevention of senile dementia (Figure 1c). On the other hand, a noncontact coded lock responding to external magnetic fields takes advantage of less risks in cross infection and personal information leakage (fingerprint

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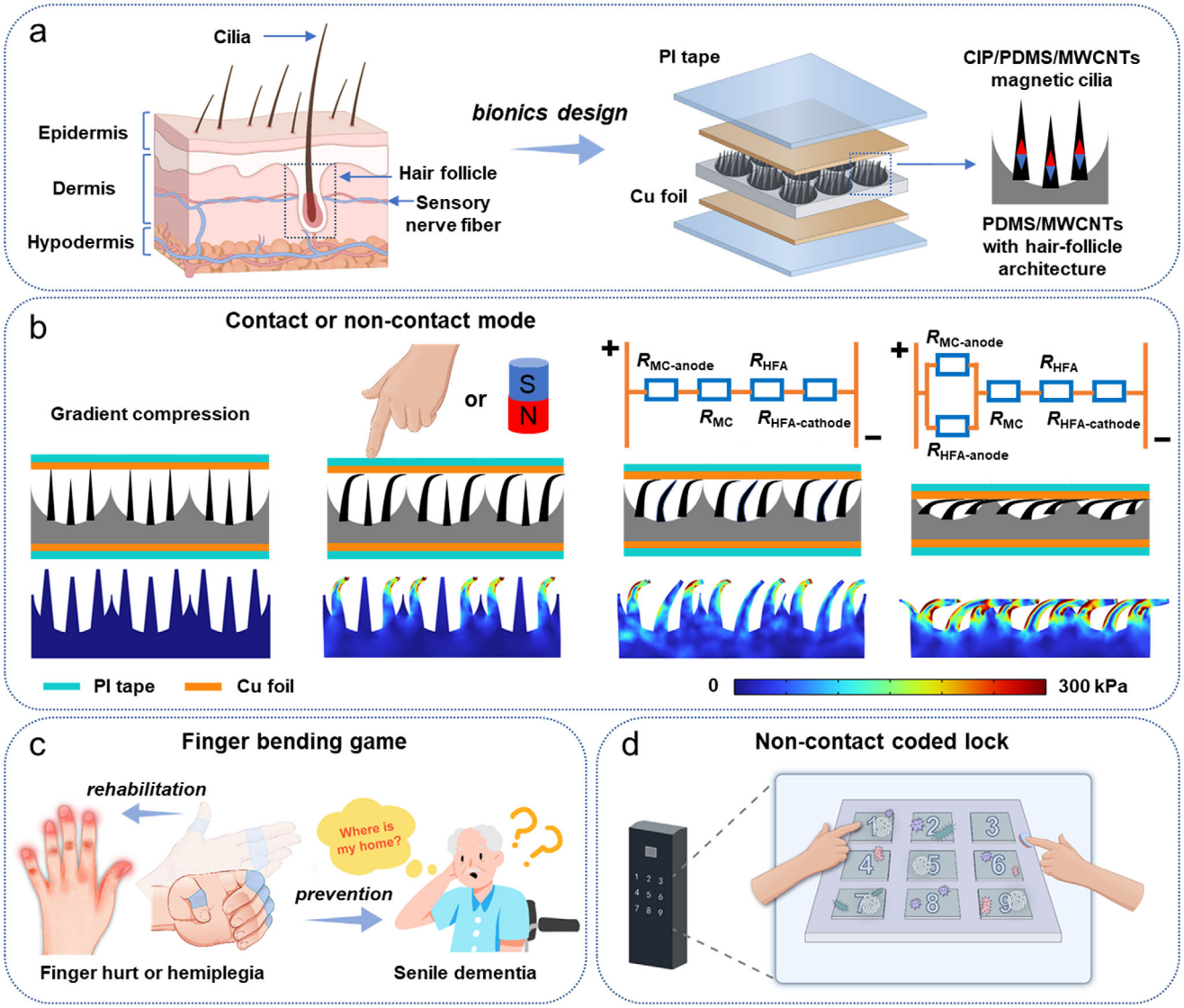

Figure 1. a) Schematic illustration of the bionics design of an MCHFA-based sensor. b) Scheme illustration, finite element analysis, and equivalent circuit analysis of the sensing mechanism of the MCHFA-based sensor in contact and non-contact mode. c) Finger bending game for finger rehabilitation training and prevention of senile dementia. d) The application in noncontact coded lock of the MCHFA-based sensor array.

for instance) compared with the traditional contact-coded locks (Figure 1d).

2.2. Sensing Performances in Contact Mode

The current variations with applied pressures were studied in detail to investigate the pressure response of the MCHFA-based sensor. The $I-V$ curves of the MCHFA-based sensor were measured at various pressures with voltage varying from −6 to 6 V, as shown in **Figure 2a**, where current is proportional to voltage, indicating a good ohmic contact between the MCHFA-based pressure sensing layer and the copper electrodes. The sensitivity, one of the most important parameters for the pressure sensors, is defined as

$$S = \left(\Delta I / I_0\right) / \Delta P \tag{3}$$

where I_0 is the initial current, ΔI is the current variation, and ΔP is the change of applied pressure. Furthermore, linearity R^2 is another key parameter to value the sensing performances. Figures S2–S4a,b, and Table S3 (Supporting Information) show a detailed study about the dependence of the cilia length (determined by the rotation speed of the spin coating) and the hair-follicle diameter D on the pressure response behavior based on the sensitivity and linearity. As a result, the MCHFA-based sensor with long cilia (rotate speed: 500 r min^{-1}) and a moderate hair-follicle diameter (D = 3 mm) possesses the best performance in the balance between sensitivity and linearity. The stress-strain curves of the MCHFA-based pressure sensing layers prepared under different rotate speeds and hair-follicle diameters also show that the one with rotate speed of 500 r min^{-1} and D = 3 mm takes advantage in high sensitivity and wide linear response range (Figure S4c, Supporting Information). Figure 2b

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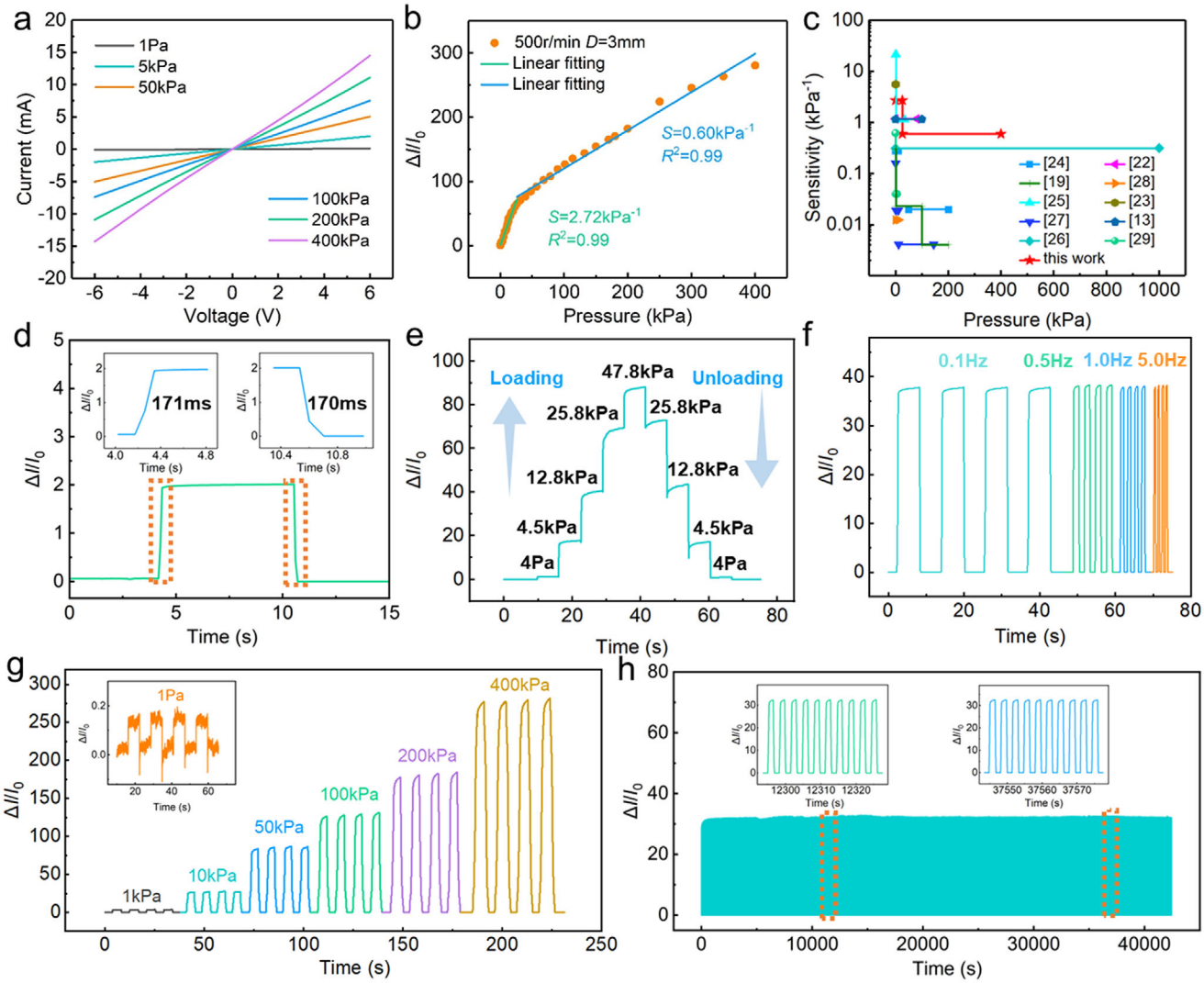

Figure 2. a) *I–V* curves of the MCHFA-based sensor under different loading pressures. b) Relative current changes of the MCHFA-based sensor over a broad pressure regime up to 400 kPa. c) Comparison of sensitivity and pressure-response range of the MCHFA-based sensor with the existing pressure sensors using magnetic fillers. d) Response and recovery time of the MCHFA-based sensor. e) Stably reversible dynamic pressure response between low (≈4 Pa) and high pressure (≈47.8 kPa). f) Pressure-response stability of the MCHFA-based sensor under various driving frequencies. g) Cyclic responses of the MCHFA-based sensor to varied pressures ranging from 1 Pa to 400 kPa. The inset shows the limit of detection as low as 1 Pa. h) Durability test of the MCHFA-based sensor up to 10 000 loading/unloading cycles at 10 kPa.

indicates that the MCHFA-based sensor possesses a wide linear response range up to 26 kPa with high sensitivity (2.72 kPa^{-1}) and linearity (R^2 = 0.99), since the continuous deformation of the cilia with different height increases the contact area between the cilia and the electrode, as illustrated in Figure 1b. Although the sensitivity reduces above 26 kPa, a much wider linear response range from 26 to 400 kPa occurs, which may be attributed to the synergistic effect of the cilia and the hair-follicle architecture under large applied pressures. Compared with the existing pressure sensors using magnetic fillers (Figure 2c), the gradient compressibility of the magnetic cilia with different height and the hair-follicle architecture endows the MCHFA-based sensor's high sensitivity and wide linear response range.[13,19,22–29]

Furthermore, the dynamical response behaviors were investigated. As exhibited in Figure 2d, the response time and recovery time of the MCHFA-based sensor are 171 and 170 ms, respectively. The MCHFA-based sensor could also track a quickly changing pressure between low pressure (4 Pa) and high pressure (47.8 kPa) (Figure 2e). During pressure loading/unloading, the response behavior is almost reversible. In addition, vibration signals at different frequencies (0.1, 0.5, 1.0, and 5.0 Hz) can be accurately detected without obvious response delay or frequency dependence due to the fast response characteristic of the MCHFA-based sensor, as shown in Figure 2f. Figure 2g exhibits the repeatable response to the cyclic loading under various pressures from 1 to 400 kPa, and even a weak pressure as low as 1 Pa can be detected. Moreover, the MCHFA-based sensor exhibits remarkable stability and durability within 10 000 loading/unloading cycles at 10 kPa (Figure 2h), and even shows good robustness under extreme conditions of high temperature (50 °C)

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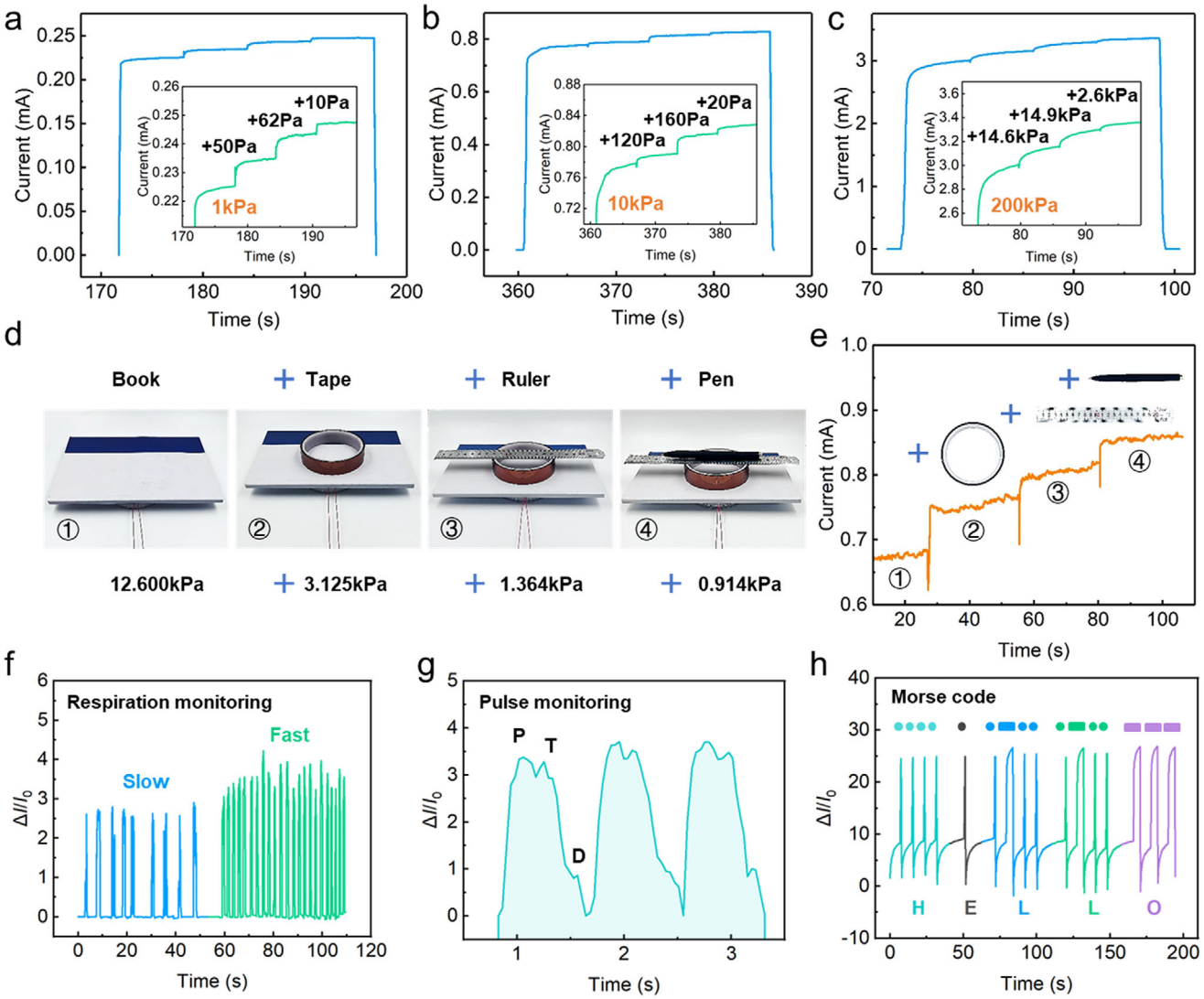

Figure 3. a–c) Detection of tiny pressure steps under loading pressure of a) 1 kPa, b) 10 kPa, and c) 200 kPa, respectively. d) Detection of tiny pressure objects successively placed on one book weighing 12.600 kPa. e) Current signals corresponding to panel (d). f) Relative current changes in respiration monitoring. g) Relative current changes in pulse monitoring. h) Morse code "HELLO" encoded by controlling the action time and interval of the finger tapping.

and high relative humidity (RH = 82%), as shown in Figure S4d,e (Supporting Information).

2.3. Excellent Pressure Resolution

Many existing piezoresistive pressure sensors tend to lose perception ability as the resolution degenerates with increasing pressure due to the limited compressibility of the matrix materials,[37–43] especially for those filled with magnetic materials.[13,19–23] However, an ideal flexible pressure sensor should possess the ability to detect tiny pressure variations not only under low pressures but also under high pressures. The synergistic effect of the cilia and the hair-follicle architecture endows the sensor a good compressibility over a wide pressure range, leading to an ultrahigh resolution of pressure perception under low, medium, and high applied pressures. Pressure discrimination performances under reference pressures of low pressure (1 kPa), medium pressure (10 kPa), and high pressure (200 kPa) were shown in **Figure 3**a–c, respectively. The MCHFA-based sensor was first compressed to the reference pressures, followed by loading three tiny pressures in sequence. Tiny pressure steps under the three reference pressures of 1, 10, and 200 kPa are clearly visible in the insets of Figure 3a–c, respectively. The corresponding pressure resolutions under reference pressures of low pressure (1 kPa), medium pressure (10 kPa), and high pressure (200 kPa) are 0.90%, 0.20%, and 1.13%, respectively, which are much better than those of human skin.[44]

Several interesting tests were carried out to further exhibit such extraordinary pressure resolution (Figure 3d–h). To uniformly load pressure to the 1 cm × 1 cm MCHFA-based sensor, we used three 1 cm × 1 cm × 1 mm glass slides to connect

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the sensor with a hard-shell book, which was set as the reference pressure of ≈12.600 kPa (Figure 3d). A tape (≈3.125 kPa), a ruler (≈1.364 kPa), and a pen (≈0.914 kPa) were gently put onto the center of the hard-shell book in sequence. The corresponding current changes are clearly exhibited in Figure 3e, which shows that each tiny pressure change can be precisely recognized with a high-pressure resolution of 5.35%. Due to the excellent pressure resolution, the MCHFA-based sensor exhibits a fine recognition of weak signals, such as the respiratory rate (Figure 3f), the pulse wave with obvious typical peaks reflecting hemodynamic details (Figure 3g), and the Morse code controlled by the finger tapping (Figure 3h), and so on. Additionally, the MCHFA-based sensor possesses a remarkable performance in human motion monitoring by sensing and recognizing the deformation details of muscles (Figure S5, Supporting Information). In terms of the performance under high-pressure conditions, some pressure sensing measurements by mounting a MCHFA-based pressure sensor on the left rear wheel of a car (1600 kg) were performed, as exhibited in Figure S6a (Supporting Information). As the car was driven forward or backward, the MCHFA-based pressure sensor presented fast response signals and even the vehicle speed could be detected (Figure S6b, Supporting Information), indicating the sensor works well under extremely high pressure up to ≈400 kPa. Furthermore, the MCHFA-based pressure sensor can recognize the pressure changes as the driver gets on or off the car (Figure S6c, Supporting Information), showing a high-pressure resolution under high-pressure conditions.

2.4. Finger Bending Training Game

Injuries to finger joints are common in daily life. Postoperative rehabilitation training is very important for the recovery of finger function. However, the existing devices for finger rehabilitation training are usually bulky, and repeated practice may be boring. To help the patients possess a convenient, comfortable, and relaxed recovery process, we designed a wearable finger rehabilitation training glove equipped with an interesting game, as shown in **Figure 4a**. The MCHFA-based sensor attached to each finger can stably detect real-time relative current variation corresponding to different angles of finger bending (Figure 4b,c). A block diagram of the hardware system of the finger bending training game is shown in Figure 4d, indicating the signal flow. The sensors' relative current variations are converted into voltage changes using a linear voltage conversion module (LVCM). An analog-digital converter (ADC) should be used to transform the obtained voltage changes from analog to digital signals, which the microcontroller unit (MCU) can use. A universal asynchronous receiver/transmitter (UART) is then used to send the signals to the application (APP). Figure 4e exhibits the flow chart of the finger bending angle recognition algorithm for the finger bending training game. First, a random target bending angle from 30^0, 60^0, and 90^0 will be generated in the APP. Second, effective time for bending finger from 0^0 to the target angles, i.e., the vertical width of grey area, can be preset as a constant or changing mode. After the setting of the target angles and effective time is finished, a rectangle containing the target angles and effective time is formed in the game interface and it is falling straight down at constant speed. Until the rectangle approaches the red line, voltage signals corresponding to the real-time finger bending angles will be acquired from the MCHFA-based sensors. If the finger can be bent to the target angle in the effective time, the game score will be plus one. Otherwise, the game score keeps unchanged.

There are three optional modes in the training game of finger bending. The first mode aims to controlling the bending angles of one finger (index for instance) within a consistent valid time (Figure 4f; Movie S1, Supporting Information), while the second mode focuses on training the reaction speed of bending a finger to a preset angle during the ever-changing valid time (Figure 4g; Movie S1, Supporting Information). Moreover, the third mode can be used to comprehensive exercises by controlling the bending angles of five fingers under various valid time (Figure 4h; Movie S1, Supporting Information), which may be considered as a deep combination of the first and second mode.

2.5. Sensing Performances in Noncontact Mode

For the non-contact mode, the MCHFA-based sensor can be used to detect the magnetic field intensity H. As shown in **Figure 5a**, when the MCHFA-based sensor is in an external magnetic field, the CIP/PDMS cilia will be bended, leading to the increase of contact area between the pressure sensing layer and the electrode, i.e., the decrease of resistance in the sensor. Figure 5b exhibits the dependence of the relative current variations of the sensor on the magnetic field intensity. The sensitivity in the non-contact mode is defined as

$$S = \left(\Delta I / I_0\right) / \Delta H \tag{4}$$

where I_0 is the initial current, ΔI is the current variation, and ΔH is the change of magnetic field intensity. The MCHFA-based sensor possesses a linear response ($R^2 = 0.94$) with a sensitivity of 2.8 T^{-1} over a wide range of magnetic field intensity from 30 to 171 mT. In addition, the limit of detection (LOD) is low to 10 mT with a response and recovery time of 148 and 147 ms (Figure 5c), respectively. Furthermore, the MCHFA-based sensor exhibits a high resolution to the magnetic field intensity, as shown in Figure 5d–f. When the MCHFA-based sensor was located in a varying magnetic field, clear steps of 10.8, 12.2, and 13.0 mT under a reference magnetic field of 50 mT, and 23, 22, and 16 mT under an initial magnetic field of 100 mT, can be seen in Figure 5e,f, respectively. The corresponding resolutions under the magnetic field intensity of 50 and 100 mT are 17.8% and 11.0%, respectively. Additionally, a stable and repeated detection between 50 and 100 mT up to 2700 cycles was observed in Figure 5g, indicating a robust characteristic of the MCHFA-based sensor in detecting the magnetic field intensity. Moreover, by comparison of the sensing performance in noncontact mode (Figure 5h), the MCHFA-based sensor shows a high-sensitivity and wide-linear-range response to the external magnetic field due to the gradient compressibility of the magnetic cilia with different heights on the hair-follicle architecture.[13,19–22,30–33]

2.6. Noncontact Coded Lock

In some special places, especially in hospitals, non-contact function buttons, including coded locks for entrance guards, lift

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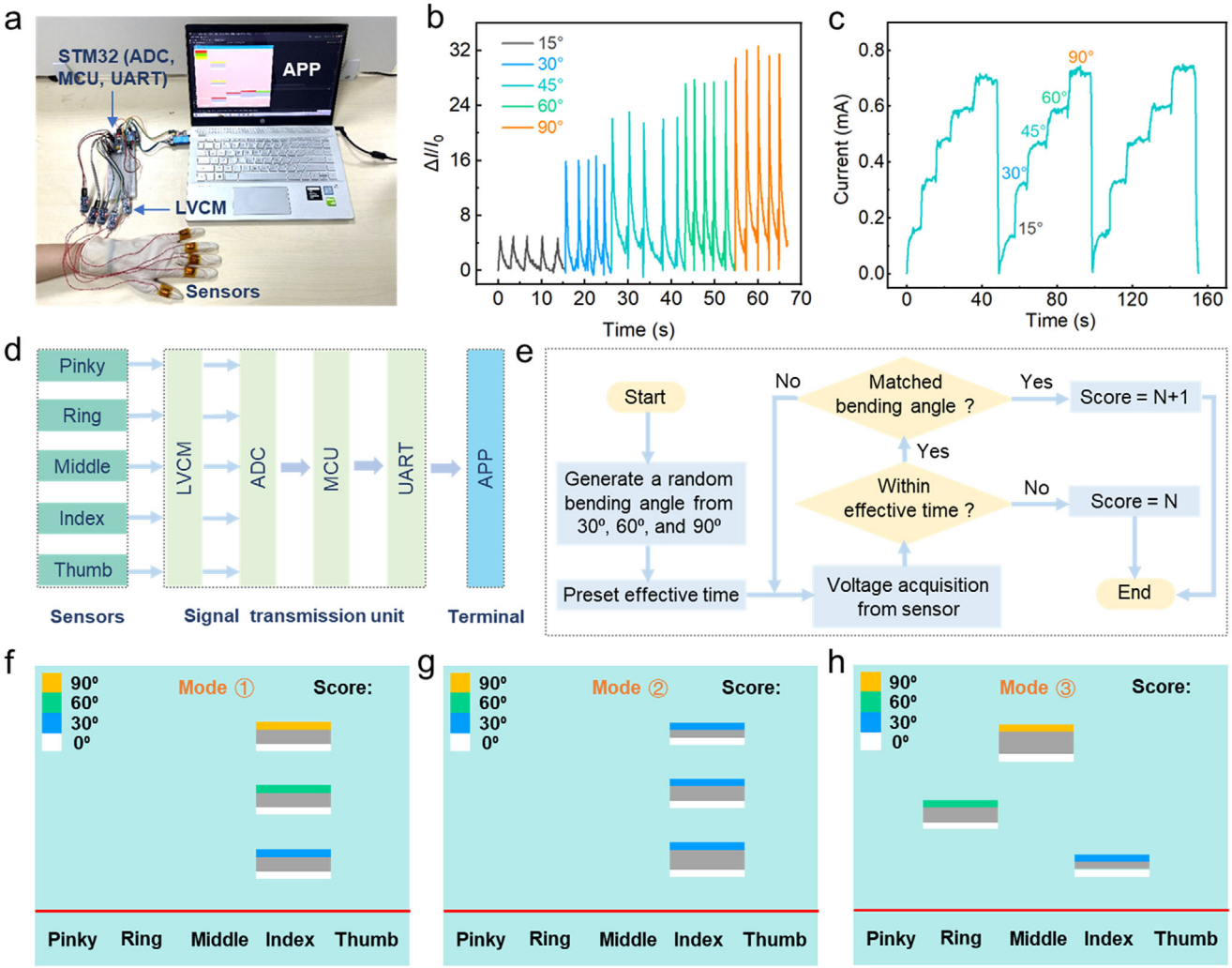

Figure 4. a) Photographs of hardware and software systems for the finger-bending training game. b) Relative current changes during the finger bending with different angles. c) Dynamical response to the finger bending angle of 15^0, 30^0, 45^0, 60^0, and 90^0 in sequence. d) Block diagram of the hardware system showing the signal flow in the finger-bending training game. e) Flow chart of the finger bending angle recognition algorithm for the finger bending training game. f) Mode 1: controlling the bending angles of one finger (index for instance) within a consistent valid time. g) Mode 2: training the reaction speed of bending a finger to a preset angle during an ever-changing valid time. h) Mode 3: comprehensive exercises by controlling the bending angles of five fingers under various valid times.

buttons, and buttons for calling systems, are essential to avoid cross-infection. On the other hand, considering personal information security, one's fingerprint left on the contact buttons may be easily copied and stolen by others. Therefore, a non-contact coded lock was designed by using a 3 × 3 MCHFA-based sensor array, as shown in **Figure 6a–c**. The schematic diagram of the non-contact coded lock is illustrated in Figure 6d,e: 1) When the finger attached a magnetic sheet is approaching the sensor coded from 1 to 9, the CIP/PDMS cilia in the sensor will be bent, leading to the decrease of resistance in the sensor. 2) The resistance variation of the sensor is transferred into voltage change by an LVCM. The obtained voltage change is an analog signal and should be changed into a digital signal using an ADC for further processing by the MCU and then transmission to the APP by a UART. 3) Until the detected voltage exceeds the threshold, the code to which the sensor belongs will be output. The output codes will be recorded as a coded string in chronological order. 4) Unless the real-time coded string is the same as the default password, the noncontact coded lock will not be unlocked. For example, as exhibited in Figure 6f–h and Movie S2 (Supporting Information), the non-contact coded lock with three kinds of default passwords "1475369," "14723698," and "1478963" preset according to the writing way of the letters "H," "D," and "U" has been unlocked successfully, respectively. Furthermore, the MCHFA-based sensor would provide a new approach of interaction means in the future as a non-contact command transmitter.

3. Conclusion

To adapt to multifunctional application scenarios, the desired pressure/magnetism dual-mode sensors should possess high sensitivity and wide linear response range in both contact and

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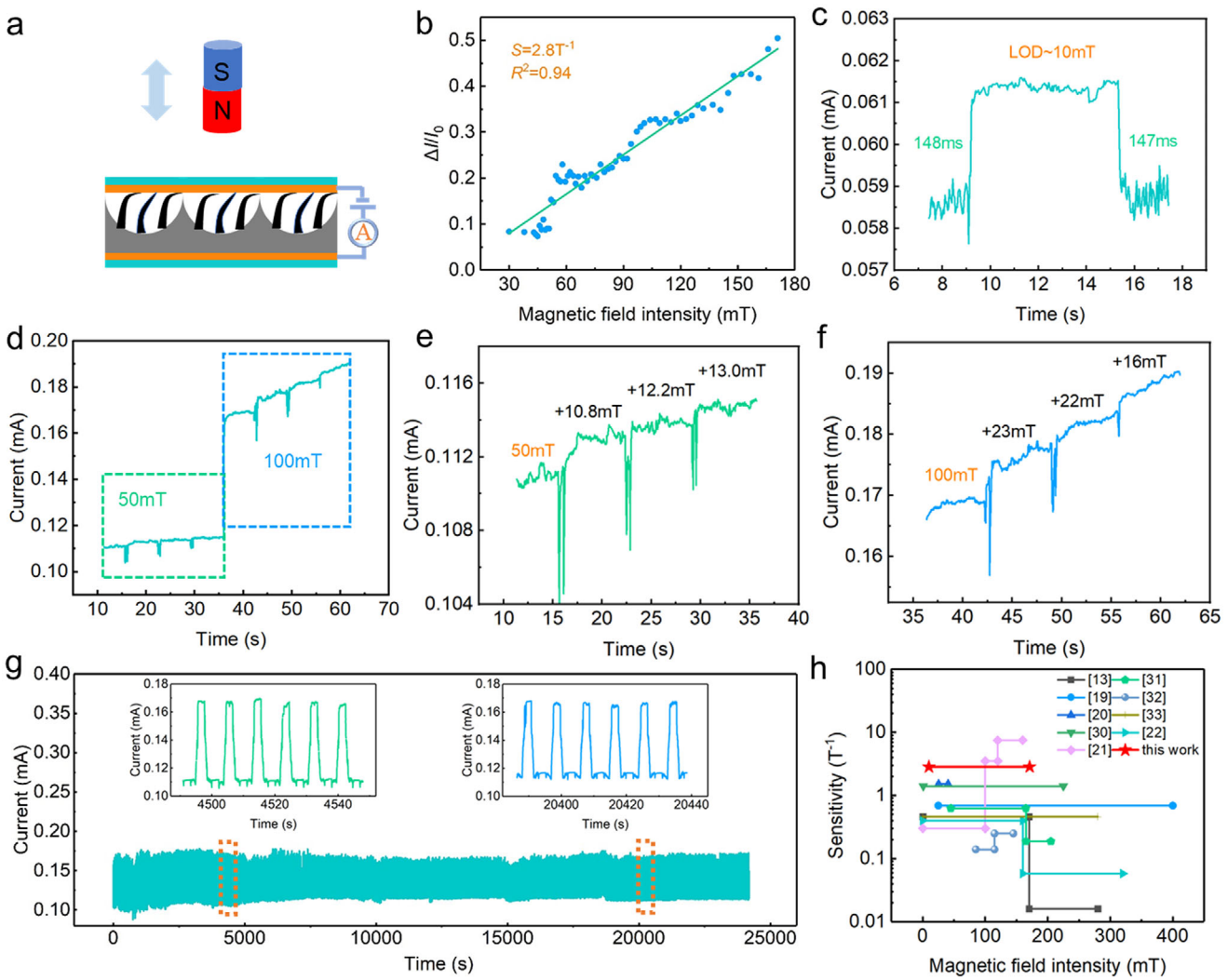

Figure 5. a) Scheme illustration of detecting an external magnetic field using an MCHFA-based sensor in noncontact mode. b) Relative current changes of the MCHFA-based sensor over a wide range of the magnetic field intensity from 30 to 171 mT. c) Limit of detection in sensing the magnetic field intensity and the corresponding response and recovery time. d) High resolution in recognizing tiny magnetic field variations under a large external magnetic field of e) 50 mT and f) 100 mT, respectively. g) Durability test of the MCHFA-based sensor up to 2700 cycles under a changing external magnetic field between 50 mT and 100 mT. h) Comparison of sensitivity and magnetic field response range of the MCHFA-based sensor with the existing pressure sensors.

noncontact mode. Here, a bionic e-skin with magnetic cilia and hair-follicle architecture is proposed to achieve high-sensitivity and wide-linear-range pressure/magnetism dual-mode perception, where the high sensitivity is mainly dominated by the easy deformation of cilia and the wide linear range is attributed to the gradient compressibility of the synergistic architectures of the cilia and the hair-follicle. In contact mode, the pressure perception shows a wide linear response range with high sensitivity and linearity (0–26 kPa: $S = 2.72$ kPa^{-1}, $R^2 = 0.99$; 26–400 kPa: $S = 0.60$ kPa^{-1}, $R^2 = 0.99$). For the non-contact mode, the magnetism response possesses a linear response ($R^2 = 0.94$) with a high sensitivity of 2.8 T^{-1} over a wide range of the magnetic field intensity from 30 to 171 mT. Additionally, high resolutions of pressure perception (0.20%) and magnetism response (11.0%) are obtained due to the wide-linear-range high sensitivity in both modes. The MCHFA-based dual-mode sensor also exhibits fast response/recovery time and remarkable stability. These excellent characteristics in dual modes endow the MCHFA-based sensor multi-functional applications in health monitoring and human-machine interfaces, such as a finger bending game for the rehabilitation training and the prevention of senile dementia, and a non-contact coded lock taking advantage of fewer risks in cross-infection and personal information leakage. Furthermore, creating such dual-mode sensors can initially eliminate the need for extra sensing equipment and connections, which is significant in terms of affordability and easier access to practical applications.

4. Experimental Section

Materials: Dow Corning Corporation (Sylgard 184, America) supplied the cross-linker and pre-polymer for the polydimethylsiloxane (PDMS).

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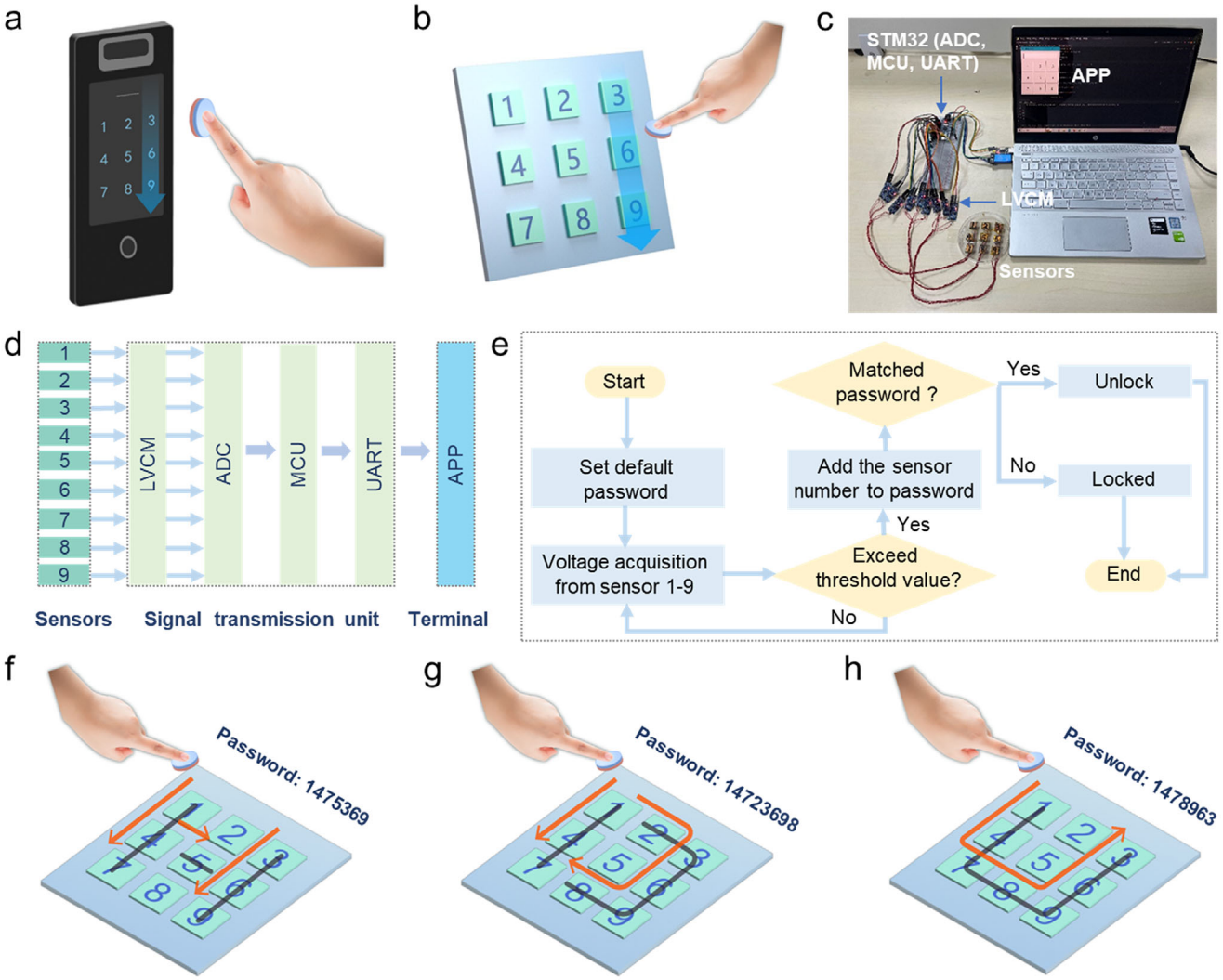

Figure 6. a) Scheme illustration of a noncontact code lock based on the MCHFA-based sensor array. b) Enlarged view of the operation interface of the noncontact code lock. c) Photographs of hardware and software systems for the non-contact code lock. d) Block diagram of the hardware system showing the signal flow in the noncontact code lock. e) Flow chart of the code recognition algorithm for the non-contact code lock. F–h) Unlock processes under three default passwords of f) "1 475 369," g) "14 723 698," and h) "1 478 963," respectively.

From Shanghai Macklin Biochemical Technology Co., Ltd. (Shanghai, China), carbonyl iron powder (CIP) was acquired. The supplier, Nanjing Xianfeng Nanomaterials Technology Co., Ltd. (Nanjing, China), provided the multi-walled carbon nanotubes (MWCNTs, diameter: 20–30 nm, length: 10–30 µm, purity >95%). Balls of stainless steel with diameters of 2.00, 3.00, and 4.00 ± 0.01 mm were bought from the neighborhood market. Polyimide (PI) tapes and copper foil electrodes were acquired from 3 M Corporation (America).

Preparation of the MCHFA-Based Sensor: The preparation process of the MCHFA-based pressure sensing layer, which was inspired by the human skin with the cilia and hair-follicle architecture, is shown in Figure S1 (Supporting Information). First, closely packed balls with their half bottom immersed in the deionized water were used as a template, which was then covered by the PDMS with a mass ratio of 10:1 between pre-polymer and cross-linker to prepare the bionic skin with hair-follicle architecture. The PDMS with hair-follicle architecture (HFA) was peeled off and shaped into a 1 cm × 1 cm piece after curing for 5 h at 70 °C. The magnetic cilia were subsequently obtained by spin coating the HFA-based PDMS piece with a CIP/PDMS mixture with a mass ratio of 1:1 and curing it for 5 h at 60 °C in a 300 mT magnetic field. After immersing the HFA-based PDMS piece with CIP/PDMS magnetic cilia in a 10 weight percent MWCNTs dispersion and swirling it for an hour at 1500 rpm, it was placed in an oven and baked for 5 h at 60 °C. Following MWCNTs coating, a piezoresistive sensor was created by sandwiching the MCHFA-based pressure sensing layer between two copper foil electrodes. PI tapes were then used to further enclose the sensor.

Characterization and Measurement: The MCHFA-based pressure sensing layer's morphology was examined using a scanning electron microscope (SEM, JSM-IT500HR, Japan) and a digital camera. During the pressure sensing test, a stepper motor (Zolix CXPF100-80175, China) coupled with a motion controller (Zolix SC300-1A, China) was used to load and unload pressure onto the sensor. A precision balance (Sartorius BSA224S-CW, Germany) was used to quantify the applied force in the contact mode. The associated pressure was then determined by dividing the force by the sensing area (1 cm^2), ignoring the Poisson effect. Regarding the non-contact mode, a NdFeB permanent magnet (diameter: 4 cm, height: 2 cm) was fixed to the stepper motor and a sensor with a size of 1 cm × 1 cm was put on the central axis area of the permanent magnet. A Gauss

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meter (TD8620, China) was used to detect the magnetic field strength on the sensor surface. This was controlled by the distance between the magnet and the sensor. The magnetic field strength of three positions on the sensor surface (A: the center of the sensor; B: the midpoint of the side length of the sensor; C: the vertex of the sensor) was measured to analyze the magnetic field distribution (Figure S7, Supporting Information). By comparison of the magnetic field strength of three positions A, B, and C for various distances between the magnet and the sensor (Table S4, Supporting Information), it is found that their values were basically consistent, i.e., the magnetic field on the sensor surface is nearly uniform. Therefore, the magnetic field strength was chosen on the center of the sensor surface as the measurement result. A Keysight B2902A meter was used to measure the sensor's real-time current variation in both modes.

Finite Element Analysis: Finite element analysis (FEA) was conducted based on the neo-Hookean model using the commercial package ANSYS 2020. For simplicity, the PI-Cu electrode was modeled as a rigid plate. Frictions during all contact interactions were ignorable and no penetration occurred.

All experiments involving the use of human subjects were approved by the Ethics Committee of Hangzhou Dianzi University. Informed consent was obtained for the experiments involving human participants.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

hair-follicle architecture, high sensitivity, magnetic cilia, pressure/magnetism perception, wide linear range

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[1] Y. Luo, M. R. Abidian, J.-H. Ahn, D. Akinwande, A. M. Andrews, M. Antonietti, Z. Bao, M. Berggren, C. A. Berkey, C. J. Bettinger, J. Chen, P. Chen, W. Cheng, X. Cheng, S.-J. Choi, A. Chortos, C. Dagdeviren, R. H. Dauskardt, C. Di, M. D. Dickey, X. Duan, A. Facchetti, Z. Fan, Y. Fang, J. Feng, X. Feng, H. Gao, W. Gao, X. Gong, C. F. Guo, et al., *ACS Nano* **2023**, *17*, 5211.
[2] C. Kaspar, B. J. Ravoo, W. G. van der Wiel, S. V. Wegner, W. H. P. Pernice, *Nature* **2021**, *594*, 345.
[3] R. Liu, Z. L. Wang, K. Fukuda, T. Someya, *Nat. Rev. Mater.* **2022**, *7*, 870.
[4] D. Zhong, C. Wu, Y. Jiang, Y. Yuan, M. Kim, Y. Nishio, C.-C. Shih, W. Wang, J.-C. Lai, X. Ji, T. Z. Gao, Y.-X. Wang, C. Xu, Y. Zheng, Z. Yu, H. Gong, N. Matsuhisa, C. Zhao, Y. Lei, D. Liu, S. Zhang, Y. Ochiai, S. Liu, S. Wei, J. B. H. Tok, Z. Bao, *Nature* **2024**, *627*, 313.
[5] D. Lei, N. Liu, T. Su, Q. Zhang, L. Wang, Z. Ren, Y. Gao, *Adv. Mater.* **2022**, *34*, 2110608.
[6] H. Niu, H. Li, S. Gao, Y. Li, X. Wei, Y. Chen, W. Yue, W. Zhou, G. Shen, *Adv. Mater.* **2022**, *34*, 2202622.
[7] S. Yu, Q. Ye, B. Yang, X. Liu, H. Zhou, L. Hu, C. Lu, *Chem. Eng. J.* **2024**, *497*, 154952.
[8] O. A. Araromi, M. A. Graule, K. L. Dorsey, S. Castellanos, J. R. Foster, W.-H. Hsu, A. E. Passy, J. J. Vlassak, J. C. Weaver, C. J. Walsh, R. J. Wood, *Nature* **2020**, *587*, 219.
[9] C. Lu, Y. Gao, X. Chan, W. Yu, H. Wang, L. Hu, L. Li, *Mater. Horiz* **2024**, *11*, 510.
[10] L. Shi, Z. Li, M. Chen, T. Zhu, L. Wu, *Adv. Mater.* **2023**, *35*, 2210091.
[11] L. Wang, X. Qi, C. Li, Y. Wang, *Adv. Funct. Mater.* **2024**, *34*, 2409358.
[12] K. R. Jinkins, S. Li, H. Arafa, H. Jeong, Y. J. Lee, C. Wu, E. Campisi, X. Ni, D. Cho, Y. Huang, J. A. Rogers, *Sci. Adv.* **2022**, *8*, abo0537.
[13] Y. Fu, C. Yang, B. Zhang, Z. Wan, S. Wang, K. Zhang, L. Yang, R. Wei, *Adv. Funct. Mater.* **2024**, *35*, 2416453.
[14] S. Li, Y. Zhang, X. Liang, H. Wang, H. Lu, M. Zhu, H. Wang, M. Zhang, X. Qiu, Y. Song, Y. Zhang, *Nat. Commun.* **2022**, *13*, 5416.
[15] L. Hu, F. Liu, Q. Quan, C. Lu, S. Yu, L. Li, *Adv. Funct. Mater.* **2024**, *34*, 2409085.
[16] H. Wang, W. Wang, J. J. Kim, C. Wang, Y. Wang, B. Wang, S. Lee, T. Yokota, T. Someya, *Sci. Adv.* **2023**, *9*, adi2445.
[17] Y. Zhou, X. Zhao, J. Xu, Y. Fang, G. Chen, Y. Song, S. Li, J. Chen, *Nat. Mater.* **2021**, *20*, 1670.
[18] X. Zhao, Y. Zhou, J. Xu, G. Chen, Y. Fang, T. Tat, X. Xiao, Y. Song, S. Li, J. Chen, *Nat. Commun.* **2021**, *12*, 6755.
[19] Q. Zhou, B. Ji, B. Hu, S. Li, Y. Xu, Y. Gao, W. Wen, J. Zhou, B. Zhou, *Nano Energy* **2020**, *78*, 105382.
[20] L. Ding, Y. Wang, C. Sun, Q. Shu, T. Hu, S. Xuan, X. Gong, *ACS Appl. Mater. Interfaces* **2020**, *12*, 20955.
[21] Y.-F. Liu, Y.-F. Fu, Y.-Q. Li, P. Huang, C.-H. Xu, N. Hu, S.-Y. Fu, *J. Mater. Chem. B* **2018**, *6*, 896.
[22] Y. Fu, S. Wang, Y. Tian, B. Zhang, Z. Zhao, Z. Wan, X. Chen, D. Zhu, L. Yang, Z. Wei, *Chem. Eng. J.* **2024**, *497*, 154441.
[23] P. Huang, D. Tan, Q.-M. Li, Y.-Q. Li, Y.-Q. Fu, N. Hu, S.-Y. Fu, *ACS Appl. Mater. Interfaces* **2020**, *12*, 8674.
[24] Q. Zhou, B. Ji, Y. Wei, B. Hu, Y. Gao, Q. Xu, J. Zhou, B. Zhou, *J. Mater. Chem. A* **2019**, *7*, 27334.
[25] J. Liu, Z. Wen, H. Lei, Z. Gao, X. Sun, *Nano-Micro Lett.* **2022**, *14*, 88.
[26] B. Ji, Q. Zhou, B. Hu, J. Zhong, J. Zhou, B. Zhou, *Adv. Mater.* **2021**, *33*, 2100859.
[27] W. Asghar, F. Li, Y. Zhou, Y. Wu, Z. Yu, S. Li, D. Tang, X. Han, J. Shang, Y. Liu, R.-W. Li, *Adv. Mater. Technol.* **2020**, *5*, 1900934.
[28] B. Ji, Q. Zhou, G. Chen, Z. Dai, S. Li, Y. Xu, Y. Gao, W. Wen, B. Zhou, *J. Mater. Chem. C* **2020**, *8*, 15634.
[29] H. Chen, Y. Jing, J.-H. Lee, D. Liu, J. Kim, S. Chen, K. Huang, X. Shen, Q. Zheng, J. Yang, S. Jeon, J.-K. Kim, *Mater. Horiz.* **2020**, *7*, 2378.
[30] Q. Zhou, B. Ji, F. Hu, J. Luo, B. Zhou, *Nano-Micro Lett.* **2021**, *13*, 197.
[31] Q. Shu, Z. Xu, S. Liu, J. Wu, H. Deng, X. Gong, S. Xuan, *Chem. Eng. J.* **2022**, *433*, 134424.
[32] L. Ding, S. Xuan, L. Pei, S. Wang, T. Hu, S. Zhang, X. Gong, *ACS Appl. Mater. Interfaces* **2018**, *10*, 30774.
[33] Y. Fu, S. Zhao, B. Zhang, Y. Tian, D. Wang, X. Ban, Y. Ma, L. Jiang, Z. Wan, Z. Wei, *Int. J. Biol. Macromol.* **2024**, *264*, 130482.

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[34] S. Wang, X. Wang, Q. Wang, S. Ma, J. Xiao, H. Liu, J. Pan, Z. Zhang, L. Zhang, *Adv. Mater.* **2023**, *35*, 2304701.
[35] H. Luo, G. Pang, K. Xu, Z. Ye, H. Yang, G. Yang, *Adv. Mater. Technol.* **2021**, *6*, 2100616.
[36] B. Zhang, Z. Xiang, S. Zhu, Q. Hu, Y. Cao, J. Zhong, Q. Zhong, B. Wang, Y. Fang, B. Hu, J. Zhou, Z. Wang, *Nano Res.* **2014**, *7*, 1488.
[37] W. Zhang, Y. Xiao, Y. Duan, N. Li, L. Wu, Y. Lou, H. Wang, Z. Peng, *ACS Appl. Mater. Inter.* **2020**, *12*, 48938.
[38] Y. Jung, J. Choi, W. Lee, J. S. Ko, I. Park, H. Cho, *Adv. Funct. Mater.* **2022**, *32*, 2201147.
[39] W. Huang, H. Li, L. Zheng, X. Lai, H. Guan, Y. Wei, H. Feng, X. Zeng, *Chem. Eng. J.* **2021**, *426*, 130837.
[40] Z. Chen, H. Zhuo, Y. Hu, H. Lai, L. Liu, L. Zhong, X. Peng, *Adv. Funct. Mater.* **2020**, *30*, 1910292.
[41] X. Guo, D. Zhou, W. Hong, D. Wang, T. Liu, D. Wang, L. Liu, S. Yu, Y. Song, S. Bai, Y. Li, Q. Hong, Y. Zhao, L. Xiang, Z. Mai, G. Xing, *Small* **2022**, *18*, 2203044.
[42] Y.-R. Kim, M. P. Kim, J. Park, Y. Lee, S. K. Ghosh, J. Kim, D. Kang, H. Ko, *ACS Appl. Mater. Interfaces* **2020**, *12*, 58403.
[43] W. Li, X. Jin, X. Han, Y. Li, W. Wang, T. Lin, Z. Zhu, *ACS Appl. Mater. Inter.* **2021**, *13*, 19211.
[44] N. Bai, L. Wang, Q. Wang, J. Deng, Y. Wang, P. Lu, J. Huang, G. Li, Y. Zhang, J. Yang, K. Xie, X. Zhao, C. F. Guo, *Nat. Commun.* **2020**, *11*, 209.

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